

INFRASTRUCTURE

System status

Live readiness across the research environment.



Honeycomb
Cowrie + custom services

Online



MQTT Broker
Mosquitto - port 1883

Online



IoT Gateway
Elevated response latency

Warning



ESP32 Nodes
3 of 4 nodes reachable

Warning



Database
Trace store connected

Online



Generator
G0 - G1 - G2 available

Online



Replay Engine
Sandbox testbed ready

Online



Total Runs
12

View experiment >



Attack Runs
7

View experiment >



Normal Runs
5

View experiment >



Active Run
1

View experiment >



Completed Runs
11

View experiment >



CONTROLLED TESTBED SIMULATION

Controlled Replay

Replay a validated Generated Behavior in the local simulated IoT Testbed.

REPLAYS

1

REPLAY TRACES

6

VALIDATED INPUT

Generation to replay

Replay Engine rejects any Generation without a passing Validation result.

Generation ID
GEN-001 - Passed - SEQ-001

RUN-001 → SESSION-001 → SEQ-001 → GEN-001

01 service_discovery

02 mqtt_connect

03 topic_discovery

04 mqtt_subscribe

05 mqtt_publish

06 disconnect

REPLAY ENGINE

Replay controls

Actions are simulated in local Application State. No hardware or network service is contacted.

Completed

Preparing

Connecting Testbed

Replaying Actions

Collecting Trace

Completed

Start Replay

Pause

Resume

Stop

Reset

Replay ID	Generation	Protocol	Target	Started	Completed
REPLAY-001	GEN-001	MQTT	ESP32-01	06 Oct, 16:19:29	06 Oct, 16:19:33

SIMULATION CONFIGURATION

Attacker environment

Select the protocol and target before starting a session.

TARGET PROTOCOL

Web
Ready

SSH
Ready

MQTT
Ready

TARGET DEVICE

ESP32-01

CURRENT RUN

SESSION ID

SOURCE

TARGET SERVICE

RUN-001

SESSION-001

simulated

MQTT Broker

Start Simulation

Pause

Resume

Stop

Clear

BEHAVIOR PRIMITIVES

Attack actions

Each click appends one event to the active session sequence.

Service Discovery
Probe exposed services

Login Attempt
Try a simulated credential

Device Discovery
Identify reachable devices

Topic Discovery
Enumerate available topics

MQTT Connect
Open a simulated broker connection

MQTT Subscribe
Subscribe to a device topic

MQTT Publish

Device Interaction

SESSION TIMELINE

Attack session

Events captured in sequence.

SESSION ID

SESSION-001

Completed

RUN ID

SOURCE IP

TARGET

SERVICE

RUN-001

198.51.100.42

ESP32-01

MQTT Broker

16:19:29

service_discovery

Discovery - MQTT

Source 198.51.100.42 Target ESP32-01

MQTT broker service discovered

16:19:29

mqtt_connect

Interaction - MQTT

Source 198.51.100.42 Target ESP32-01

Broker connection established

16:19:29

topic_discovery

Enumeration - MQTT

Source 198.51.100.42 Target ESP32-01

Device topics enumerated

16:19:29

mqtt_publish

Manipulation - MQTT

Source 198.51.100.42 Target ESP32-01

Publish payload captured

NEW RUN CONFIGURATION

Create experiment

Run IDs are assigned from the shared experiment store.

Run ID

Scenario ID *

RUN-002

SCN-MQTT-01

Assigned automatically to avoid duplicate runs.

Scenario Name *

Target Service *

MQTT Abuse

MQTT Broker

Attack Type *

Duration *

MQTT

5 minutes

Target Device *

ESP32-01

Description *

Observe MQTT topic discovery and unauthorized publish behavior.

Required fields *

Create Experiment

SELECTED RUN

Run detail

Metrics and lifecycle status for the selected experiment.

SCN-MQTT-01

RUN-001

MQTT Abuse

SCENARIO

START TIME

MQTT Abuse

06 Oct 2026, 16:19 ICT

DURATION

TARGET

5 minutes

ESP32-01 - MQTT Broker

EVENT COUNT

PACKET COUNT

SESSION COUNT

5

5

1

RUN LIFECYCLE

Preparing

Running

Collecting Trace

Processing

Completed

Start Experiment

Stop Experiment

Reset Experiment

Related sessions

View traces

TRACE-DRIVEN GENERATION

Behavior Generator

Generate related behaviors from an extracted sequence with full Run and Session traceability.

INPUT SEQUENCES

1

GENERATIONS

1

SOURCE SEQUENCE

Choose the observed behavior

Observed Input

Input selectors stay linked to the sequence's original Run and Session.

RUN

SESSION

BEHAVIORAL SEQUENCE

LINEAGE

RUN-001

SESSION-001

SEQ-001

RUN-001 → SESSION-001 → SEQ-001

ORIGINAL SEQUENCE → SEQ-001

service_discovery → mqtt_connect → topic_discovery → mqtt_publish → disconnect

5 observed events

GENERATOR MODE

Choose a generation strategy

☒ G0 Exact Replay

Preserve each observed action in order.

☐ G1 Rule-Based Mutation

Apply explicit action, target and timing mutations.

☐ G2 Probabilistic Generation

Sample only observed transitions using their frequency.

GENERATION PREVIEW

G0 Exact Replay

Output remains traceable to SEQ-001.

ORIGINAL

service_discovery → mqtt_connect → topic_discovery → mqtt_publish → disconnect

GENERATED

STEP 01 - 0 HS - ESP32-01	service_discovery
STEP 02 - 0 HS - ESP32-01	mqtt_connect
STEP 03 - 0 HS - ESP32-01	topic_discovery
STEP 04 - 0 HS - ESP32-01	mqtt_publish
STEP 05 - 0 HS - ESP32-01	disconnect

Constraint Validator

Check generated sequences against behavior constraints.

MODULE

09 / 14

SEMANTIC CONSTRAINTS

Constraint Validator

Check generated action ordering before a behavior can enter Controlled Replay.

ACTIVE RULES

3

GENERATIONS

1

SEMANTIC RULES

Constraint Rules

RULE ID	CONDITION	REQUIRED ACTION	STATUS
CR-001	mqtt_publish	mqtt_connect	Active
CR-002	mqtt_subscribe	mqtt_connect	Active
CR-003	device_control	device_discovery	Active

Evaluation

Assess validity, similarity, diversity, and novelty.

TRACE-DRIVEN EVALUATION

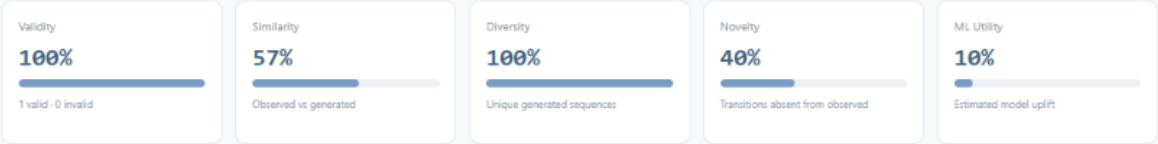
Generated behavior quality

Metrics are computed from observed events, linked behavioral sequences, generated actions and replay traces.

OBSERVED EVENTS
5

GENERATED ACTIONS
6

REPLAY TRACE ROWS
6



VALIDATION COHORT

Generated sequence validity

Constraint results are read from the Validator records.

Evaluate traces

100%

Valid generated sequences

1 passed validation

0%

Invalid generated sequences

0 rejected - 0 pending

● Observed trace

● Generated behavior

● Replay trace

